

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Graphs of Functions

Read a graph by its shape, intercepts, intersections, and domain.

GRAPHS OF FUNCTIONS · P1 · 06

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 SHAPE: Know The Parent Shape

TRIGGER *sketch a standard function*

02 ZERO: Find Roots And Intercepts

TRIGGER *where the graph crosses axes*

03 MEET: Intersections

TRIGGER *two graphs meet*

04 ASYM: Asymptotes And Domain

TRIGGER *reciprocal, square root, denominator zero*

BANK EVIDENCE Local pages: 25 problem, 47 answer, 25 notes. Website primary entries: 18.

01 SHAPE: Know The Parent Shape

TRIGGER *sketch a standard function*

BECOMES Identify the parent curve before plotting details.

FIRST LINE TO WRITE

$$y = x^2, \quad y = x^3, \quad y = \frac{1}{x}, \quad y = \sqrt{x}$$

SIMPLEST STRATEGY

- 1 Select the parent graph.
- 2 Highlight symmetry/asymptotes.
- 3 Add intercepts.
- 4 Plot key points.
- 5 End with the correct curve shape.

WORKED MODEL

$$y = (x - 2)^2$$

is the x^2 shape shifted right by 2.

HARD VARIANTS

- 1 Transform the parent graph before plotting extra points.
- 2 Mark asymptotes/endpoints before drawing reciprocal or root curves.
- 3 Use symmetry to reduce work, but check the domain.

— BOTTOM LINE

The parent shape is the memory anchor.

02 ZERO: Find Roots And Intercepts

TRIGGER *where the graph crosses axes*

BECOMES Set $y = 0$ for roots and $x = 0$ for y-intercept.

FIRST LINE TO WRITE

$$y = 0 \quad \text{or} \quad x = 0$$

SIMPLEST STRATEGY

- 1 Zero the correct variable.
- 2 Equation solve.
- 3 Record coordinates.
- 4 Order points clearly.

WORKED MODEL

| For $y = x^2 - 9$, roots: $x = \pm 3$, y-intercept: -9 .

HARD VARIANTS

- 1 If factorisation is hard, use a substitution or formula before writing intercepts.
- 2 Intercepts are coordinates; include $y = 0$ or $x = 0$ correctly.
- 3 Check whether any intercept is outside the stated domain.

— BOTTOM LINE

Intercepts are coordinates, not just numbers.

03 MEET: Intersections

TRIGGER *two graphs meet*

BECOMES Equate the two formulas.

FIRST LINE TO WRITE

$$f(x) = g(x)$$

SIMPLEST STRATEGY

- 1 Make the two expressions equal.
- 2 Expand/rearrange.
- 3 Evaluate x .
- 4 Turn each x into a coordinate.

WORKED MODEL

$$x^2 = x + 2 \Rightarrow x = 2, -1.$$

HARD VARIANTS

- 1 Equating two functions may create a quadratic or cubic; solve all valid roots.
- 2 For tangency, look for one repeated intersection.
- 3 Use either graph to find y , then verify with the other.

— BOTTOM LINE

Meeting points need both x and y .

SAME IDEA Use MEET: Intersections.

QUESTION

Find where $y = x^2$ meets $y = 2x + 3$.

YOUR SOLUTION

04 ASYM: Asymptotes And Domain

TRIGGER *reciprocal, square root, denominator zero*

BECOMES Find forbidden or endpoint values first.

FIRST LINE TO WRITE

denominator $\neq 0$, inside root ≥ 0

SIMPLEST STRATEGY

- 1 Ask what values are forbidden.
- 2 Set denominator/root condition.
- 3 Yield domain/asymptote.
- 4 Mark it before sketching.

WORKED MODEL

$y = \frac{1}{x-3}$
has vertical asymptote $x = 3$.

HARD VARIANTS

- 1 Denominator zeros give vertical asymptotes unless the factor cancels first.
- 2 Square-root domains come from inside root ≥ 0 .
- 3 Horizontal shifts move asymptotes and endpoints too.

— BOTTOM LINE

Domain restrictions shape the whole graph.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
sketch a standard function	$y = x^2, \quad y = x^3, \quad y = \frac{1}{x}, \quad y = \sqrt{x}$
where the graph crosses axes	$y = 0 \quad \text{or} \quad x = 0$
two graphs meet	$f(x) = g(x)$
reciprocal, square root, denominator zero	denominator $\neq 0, \quad$ inside root ≥ 0